ASHWANI KUMAR DWIVEDI

Software Engineer | Mobile

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SUMMARY

Software Engineer with **5.5 years** of experience designing and evolving **Android platform framework** components powering millions of **flagship devices.** Expertise in **large-scale system architecture, platform performance optimization, and reliability engineering.** Proven track record of leading **cross-functional technical initiatives, driving architectural decisions, mentoring engineers**, and **delivering scalable solutions** for complex system-level challenges.

PROJECTS

Automated Algo Trading System | Python, Docker, Linux

Developed and built a **fully computerized trading system for Indian equity markets** with reliable order execution and risk management.

- Developed modular architecture including **strategy engine, order management system, and portfolio tracking**.
- Implemented **automated risk management** including stop-loss, target exits, and position sizing limits.
- Built **fault-tolerant deployment using Docker and Docker Compose** for unattended execution.
- Implemented **real-time monitoring and alerting system** for execution failures and risk events.
- Designed **extensible strategy engine** supporting **multiple pluggable trading strategies.**

Sky Segmentation using Deep Learning | CNN, Python, Computer Vision

Created an **ML inference pipeline** for **mobile devices**, improving **image processing performance** by **30%** through latency and memory optimizations.

- **Engineered and trained a compact CNN-based semantic segmentation model**, reducing model footprint by **40% (down to <5MB)** while maintaining a **90% mIoU accuracy score** for resource-constrained hardware.
- **Established a systematized CI/CD deployment workflow** (Training → Export → AAR Build → Integration Testing), cutting cross-team integration overhead by **50%** and accelerating release cycles.
- **Optimized real-time inference performance** via INT8 quantization and memory profiling, lowering overall RAM footprint by **35%** and achieving a steady **60 FPS on mid-range devices**.

iOS Health Application | SwiftUI, JavaScript

Architected and integrated hybrid **Web-to-Native iOS** features using **SwiftUI**, custom **JavaScript** bridging, and **deep-linking pipelines** for the Health application.

- **Engineered a high-performance WebView module** in SwiftUI with backward compatibility for older iOS versions, maintaining a smooth **60 FPS rendering speed** and clean architecture.
- **Applied secure JavaScript bridging and deep-linking protocols** to enable seamless bidirectional communication and direct navigation between web components and native screens, reducing data latency by **30%**.
- **Optimized hybrid view performance and memory utilization**, cutting frame drops by **25%** and improving responsiveness across lower-end devices.
- **Standardized web-to-native integration workflows**, accelerating feature deployment cycles by **20%** for the iOS Health platform.

EDUCATION

Masters of Technology

Birla Institute of Technology Pilani

04/2022 - 03/2024 Pilani

Bachelors of Technology

National Institute of Technology Sikkim

08/2016 - 03/2020 Sikkim

EXPERIENCE

Software Engineer

Samsung R&D Institute India

02/2021 - Present Noida

Designed and maintained **Android framework components** across multiple **flagship** device lines, improving **modularity** and **maintainability.**

- Contributed **400+ production change lists** across Android framework modules used in millions of Samsung devices.
- Led stability initiatives for core Android framework modules, resolving **500+ production issues** and reducing post-release defects by **30%.**
- Architected automated **testing infrastructure spanning unit, functional, and integration testing**, reducing CI execution time by approximately **50%** while improving release confidence and platform reliability.
- Participated in design and architecture discussions for **20+ platform features**, focusing on backward compatibility, performance, and maintainability.
- Established engineering best practices adopted across a team of **15 engineers**, conducting **400+ technical code reviews** that improved code quality, architectural consistency, and long-term maintainability.
- Created **architecture documentation** and **Confluence knowledge base** for **Android framework modules**, **improving onboarding efficiency** and long-term **knowledge sharing** across engineering teams.

SKILLS

Java **C++** **Python** **Kotlin**

Android **Jetpack Compose**

Android SDK **Machine Learning**

Artificial Intelligence

MVVM **MVC** **Data Structures**

Algorithms **OOP** **Design Patterns**

System Design

Agile **Scrum** **Project Management**

Leadership **Communication**

Attention to detail **Creativity**